Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- Convex Optimization
  Reading • 8 min
- **Linear Programming**
  Reading • 15 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Linear Programming

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Linear programming (LP) is a specific class of convex optimization problems where both the objective function and the constraints are linear. Linear programming is widely used in various fields, including operations research, economics, engineering, and the military, due to its simplicity and the efficiency with which solutions can be found (Frontline Systems, n.d.) (10 minutes)

A linear programming problem is typically formulated as follows:

$$\begin{aligned} &Minimize \\ &c^{\mathrm{T}}x \\ &\text{Subject to:} \\ &Ax \leq b \\ &x \geq 0 \end{aligned}$$

Here:

- $x \in R^n$ is the vector of decision variables that we need to determine.
- $c \in R^n$ is the coefficient vector for the objective function.
- $A \in R^{m \times n}$ is the matrix representing the coefficients of the constraints.
- $b \in R^m$ is the vector representing the right-hand side of the constraints.
- $c^{\mathrm{T}}x$ represents the objective function, which is a linear function of the decision variables.

The goal of linear programming is to find the vector x that minimizes the objective function $c^T x$ while satisfying the constraints $Ax \leq b$ and $x \geq 0$.

Lagrangian Function

To solve this problem using the method of Lagrange multipliers, we introduce a vector of non-negative Lagrange multipliers $\lambda \in R^m$, one for each constraint. The Lagrangian function for the primal problem is then given by:

$$L(x, \lambda) = c^{\mathrm{T}}x + \lambda^{\mathrm{T}}(Ax - b)$$

Expanding this, we get:

$$L(x, \lambda) = \left(c + A^{\mathrm{T}}\lambda\right)^{\mathrm{T}} x - \lambda^{\mathrm{T}}b$$

Dual Problem

To derive the dual problem, we take the derivative of the Lagrangian $L(x, \lambda)$ with respect to x and set it to zero. This gives the condition for optimality:

$$c + A^{\mathrm{T}}\lambda = 0$$

This equation must be satisfied for the Lagrangian to have a stationary point. Substituting this condition into the Lagrangian, we can express the dual objective function $D(\lambda)$ as:

$$D(\lambda) = -\lambda^{\mathrm{T}}b$$

The dual problem is then to maximize this dual objective function subject to the constraints derived from the Lagrange multipliers:

$$\text{Maximize} - b^{\mathrm{T}}\lambda \text{ subject to } A^{\mathrm{T}}\lambda = -c, \lambda \geq 0$$

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